Vedarth Harikrishnan

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Summary

BEng Aeronautics and Astronautics graduate from the University of Southampton with a strong analytical and technical foundation spanning mathematics, physics, engineering and programming. Experienced in collaborative projects, computational modelling, problem-solving and team leadership through Formula Student, UAV development and society presidency. Combines quantitative reasoning with strong communication and organisational skills, seeking to contribute to challenging problems across technology, consulting and engineering.

Education

BEng Aeronautics and Astronautics, University of Southampton — Sep 2022 – Jul 2026

- **Key Modules:** Aerodynamics and Aircraft Design, Propulsion and Thermodynamics, Materials and Structures, Digital Aerospace (CFD and Finite Element Analysis), Electronics, Control and Response Systems.

Langley Grammar School, Sixth Form — Sep 2020 – Jul 2022

- **A-Levels:** Mathematics (A*), Further Mathematics (A*), Physics (A*)
- **AS-Levels:** Chemistry (A)

Skills

AI & Cloud Development: Microsoft Foundry, Agentic AI development and orchestration, RAG pipelines, MCP integration, computer vision, natural language processing and document intelligence.
Engineering Software: SolidWorks, ANSYS Fluent, ANSYS Mechanical (APDL), ANSYS Workbench, OpenVSP, XFLR5, XFOIL, Tecplot 360, Granta Edupack, Blender.
Programming: Python, Linux, Cantera, Arduino IDE, MATLAB, Simulink, HTML, CSS, Javascript.
Finance Software: VT, IRIS Elements.

Awards & Certifications

Microsoft Azure AI-103: Azure AI Apps and Agents Developer Associate.
UKMT: Gold Award and Kangaroo qualification for five consecutive years; First Place in School three times (Junior, Intermediate and Senior).
Music: ABRSM Grade 7 Piano, Grade 8 Singing, Grade 6 Music Theory; PRSSV Grade 6 Singing.
Lifeguarding: RLSS UK National Pool Lifeguard Qualification.

Projects

Fixed Wing UAV Project – Partnership with UoS and Soton UAV — Jan 2024 – Jun 2024

- Led the design and construction of a fixed-wing UAV with a focus on aerodynamic stability, maximizing maneuverability and speed range.
- Achieved a score 60% higher than average in bending resistance while using sustainable materials to minimize environmental impact.
- Designed and constructed a dual-axis stabilizing gimbal fit for the UAV with an emphasis on airworthiness.
- Successfully flight-tested the UAV, achieving speeds between 9 km/h and 30 km/h with excellent control and maneuverability.
- Developed code for a dual-axis stabilization system using sensor data from an MPU6050 accelerometer, executed by an Arduino microcontroller.

University Formula Student – Vehicle Performance Division — Sep 2023 - Jun 2024

- Developed Python code to generate and analyze Kármán-Trefftz/NACA aerofoils using conformal mapping and doublet panel methods.
- Conducted wind tunnel experiments to analyze drag, lift, and downforce coefficients.
- Performed XFLR5 and XFOIL analysis on aerofoils to obtain accurate data and evaluate aerodynamic performance.
- Collaborated with team members to integrate aero findings into the overall vehicle design.

Inviscid Suppression of Vortex-Induced Lift - Individual Project with UoS — Sep 2025 - May 2026

- Conducted high-fidelity inviscid CFD simulations using the CANARD solver via the IridisX HPC cluster.
- Simulated flat-plate aerofoils at Mach 0.1 and injected a prescribed vortex to model the lift fluctuation caused by

impact at the leading edge.

- Investigated the effect of sinusoidal wavy leading-edge geometry on the effectiveness of lift fluctuation suppression and studied the mechanism of phase cancellation.
- Developed automated Python scripts to batch-process surface pressure data from Tecplot 360, computing continuous transient lift histories via numerical methods.
- Results demonstrated that optimal WLE geometries can reliably disrupt vortex coherence, and the study achieved a 42.6% reduction in lift fluctuations.

Work Experience

Lifeguard, – Everyone Active, Langley Aug 2021 – Sep 2023

- Ensured the safety of all patrons by actively monitoring pool activities and responding promptly to emergencies.
- Certified RLSS NPLQ Lifeguard, demonstrating expertise in emergency response, spinal injury management, and aquatic first aid.
- Maintained lifeguard equipment, including rescue tubes, spinal boards, and automated external defibrillators (AED), to ensure operational readiness.
- Delivered exceptional customer service, fostering a welcoming and safe environment for all visitors.

Bicycle Repair and Maintenance, Urban Unity – Slough Jun 2022 - Sep 2022

- Developed experience diagnosing and repairing a variety of issues including drivetrain problems, brake malfunctions and wheel misalignment, ensuring optimal performance and safety.
- Gained hands-on experience using specialized tools for replacing components, gear indexing, and chain servicing.
- Contributed to community engagement by organizing workshops and events with local schools and charities to teach basic maintenance skills and restore children's bicycles.

Private Piano Instructor, – Self-Employed Nov 2021 – Present

- Taught piano to students in my local community, developing tailored lesson plans to nurture musical talent.
- Prepared students for ABRSM practical exams, achieving strong results.

Accounting and Finance Intern, Rithrish Accountancy – Slough Jul 2024 - Sep 2024

- Gained practical experience in accounting and finance, focusing on financial reporting, auditing, and tax preparation.
- Utilized financial software including VT and IRIS Elements to assist in the preparation of financial statements, budgeting, and forecasting.
- Aided in the preparation of VAT returns and corporate tax filings, ensuring compliance with UK tax regulations.
- Developed analytical and problem-solving skills by reviewing client financial data and preparing detailed reports.

President, – Hindu Society, University of Southampton Jul 2024 – Jun 2025

- Led a team of executives to organize large-scale cultural events, including the annual Diya Ball attended by over 400 guests.
- Oversaw financial management, budgeting, and sponsorship acquisition, ensuring fiscal responsibility and event success.
- Coordinated with university societies and external vendors to deliver high-quality events and initiatives.
- Managed and mentored subcommittees in marketing, events planning, sports, and learning.

Tutor (A-Level Maths, Physics, and Further Maths), MyTutor – Online Dec 2024 – Apr 2025

- Provided tailored tutoring sessions to A-Level students, achieving significant improvements in grades and confidence.
- Simplified complex topics, ensuring clarity and understanding for students of varying academic levels. Developed personalized study plans and resources to address individual student needs.